Forensic Reconstruction of Road Collisions from Independent Onboard Recordings

Lorenzo Bacchiani, Andrea Bedei, Mario Bravetti, Roberto Girau, and Giovanni Pau

***Abstract*—Reconstructing the interactions preceding a road collision is difficult when recordings have independent clocks, partial views and unrelated local track identifiers. We present a retrospective method that combines onboard radar, vehicle motion, controls and impact reports into a shared forensic chronology. The method estimates local motion through Kalman filtering and smoothing, then recovers clock offsets from contact correspondences or observed motion. Geometric association connects observations across recorders to reconstruct vehicles without their own logs and allows later evidence to resolve earlier identities. Semantic conversion relates manoeuvres and conflicts to recorded responses and contacts, preserving source observations and the provenance of identity decisions. Evaluation on 19 CARLA scenarios comprises 22 runs and 1816 reconstruction jobs with positioning and radar perturbations, including asymmetric degradation. In the reference condition, all 26 collision participant sets are correct and all 45 observable non-recording vehicles are represented. With nominal radar errors, the method preserves complete collision identification and local conflict precision of 0.954. When extreme radar corruption affects every recorder, 95.4% of collision participant sets remain correct, while local conflict precision falls to 0.313. The resulting chronology provides an auditable evidence base for examining competing accounts of a collision.**

***Index Terms*—Partial observability, collision reconstruction, automotive radar, multi-vehicle fusion, forensic analysis.**

I. Introduction

Disputed accounts of a crash are a practical problem in collision investigation. Practitioner guidance describes lane-change and multi-vehicle cases in which competing driver accounts must be checked against witness statements, vehicle data and scene evidence [1], [2]. Establishing what happened requires recovering the interactions that preceded the impact. Consider a common rear-end collision on a multilane road (Fig. 1). A travels faster in the right lane; B is slightly ahead in the lane to its left. B moves into A's lane. A brakes, but strikes B from behind. *Who caused the crash?* A could say B cut in so close that the collision could not be avoided. B could say there was room and A failed to slow in time. Both accounts fit the scene after the crash.

The disagreement concerns the few seconds before impact, which the collision itself does not preserve. B might have crossed just ahead of A, leaving almost no room even if A braked immediately. Or B might have completed the merge with a comfortable gap that A then closed while responding late, perhaps because the driver was distracted. A narrow gap and delayed braking could also combine, each using up part of the distance available before contact. The drivers may agree that B changed lanes and A braked, yet still disagree on whether A ever had room to avoid the crash. All these sequences end with A striking B from behind. Damage and final vehicle positions do not reveal the original gap, how quickly it closed, or when A began to brake. The collision is undisputed; what made it happen remains uncertain.

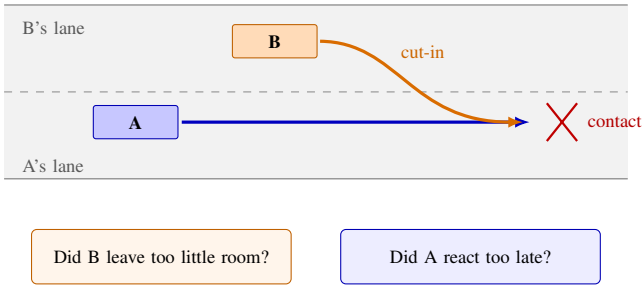


Fig. 1. A cut-in followed by a collision. The outcome alone does not distinguish an insufficient gap from a delayed response; both may occur.

Onboard recordings can preserve parts of this missing history. Their forensic value has been demonstrated through sensor-processing pipelines and reconstruction from vehicle logs and dashcams [3], [4]. Combining independent recordings, however, requires recovering correspondences in both time and physical identity. Unknown clock offsets complicate temporal alignment, while radar returns do not name the vehicles that produced them and different observers see different surfaces of the same body. Occlusion can interrupt tracking, and impacts abruptly change motion; a relevant road user may provide no recording at all. A shared chronology must therefore connect physical vehicles across observers and relate their observed interactions to recorded responses.

We present a method for retrospective forensic reconstruction from independently collected onboard recordings. Reconstruction begins by processing each recorder independently, using Kalman filtering and RTS smoothing to estimate radar-track motion and derive local events. To combine these histories, shared contacts are checked against geometry and motion to anchor a common clock; observed motion can also align recorders without a shared contact. Association and body geometry then connect different views of recording and non-recording road users, allowing later evidence to resolve earlier identities while preserving the original local observations and event times. The resulting semantic chronology relates path entries, cut-ins and conflicts to recorded controls and contacts, retaining each recorder's provenance. It provides evidence for examining competing accounts, while judgments about driver awareness and responsibility remain with the investigator.

The contributions are:

- A reconstruction method that recovers clock offsets and

L. Bacchiani, A. Bedei, M. Bravetti, R. Girau, and G. Pau are with the University of Bologna, Bologna, Italy.

physical vehicle associations across independent radar and vehicle recordings, using common-body evidence and retrospective identity resolution to reconstruct road users with and without their own logs.
- A semantic chronology that connects manoeuvres and conflict onsets to recorded responses and contacts, preserving the source observations and the provenance of later identity decisions.
- A reproducible evaluation of 19 scenarios, 22 runs and 1816 reconstruction jobs with paired positioning and radar errors, including asymmetric degradation, measuring collision identification, association consistency, geometry and local conflict recovery separately.

The forensic reconstruction code and experimental campaign are publicly available [5].

The paper is organised as follows. Section II introduces estimation, tracking and conflict measures. Section III describes the method, and Section IV evaluates its reconstruction quality and robustness. Section V discusses related research, Section VI examines threats to validity, and Section VII concludes.

II. Background

A. State Estimation: Filtering and Smoothing

State estimation infers the motion of an object from noisy observations. For planar vehicle motion, a common state contains position and velocity, $\mathbf{x}_k = (p_x, p_y, v_x, v_y)^\mathsf{T}$, at sampling instant k. For Cartesian position and directional velocity measurements, a linear state-space model describes its evolution and observation as

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{w}_k, \qquad \mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k, \tag{1}$$

where $\mathbf{F}_k$ models motion, $\mathbf{H}_k$ relates the state to the measurements, and $\mathbf{w}_k$ and $\mathbf{v}_k$ represent process and measurement noise, with covariances $\mathbf{Q}_k$ and $\mathbf{R}_k$. In a constant-velocity model, position advances by velocity times the sampling interval; process noise accounts for departures from that prediction, including acceleration [6].

The Kalman filter recursively propagates a state estimate and its error covariance, then updates them using the next measurement and its uncertainty [7]. The observations available to the estimator distinguish filtering from smoothing: filtering estimates the state at k using observations through k, whereas fixed-interval smoothing estimates that same state using the entire observation sequence, including later measurements. The Rauch–Tung–Striebel (RTS) smoother achieves this through a backward recursion over the filtered estimates [8]. In retrospective analysis, the complete recording is available, making this distinction relevant. Both estimates still depend on the assumed motion model, which must be revisited when an impact causes an abrupt velocity change.

B. Multi-Object Tracking and Data Association

Multi-object tracking estimates several trajectories while maintaining their identities over time. This is difficult because sensor detections generally carry no reliable object identity: a vehicle can produce several returns, disappear from view, or pass close to another vehicle. Data association therefore determines which observations belong to which tracks, while track management handles their initiation, continuation, and termination [6]. For a predicted state $\hat{\mathbf{x}}$ with covariance $\mathbf{P}$, the measurement innovation $\mathbf{r} = \mathbf{z} - \mathbf{H}\hat{\mathbf{x}}$ has covariance $\mathbf{S} = \mathbf{H}\mathbf{P}\mathbf{H}^\mathsf{T} + \mathbf{R}$. Compatibility can be assessed using the squared Mahalanobis distance,

$$d^2 = \mathbf{r}^\mathsf{T} \mathbf{S}^{-1} \mathbf{r}. \tag{2}$$

A statistical gate uses this distance to reject incompatible candidates. Several tracks may nevertheless explain the same observation, so passing the gate establishes compatibility without uniquely determining the association.

Automotive radar also requires an extended-object view of a vehicle. Multiple detections can arise from spatially separated reflection centres, and their distribution depends on the visible parts of the body [9]. Doppler measures relative velocity along each return's line of sight; a single return does not directly determine the tangential component. Consequently, a tracked radar position need not coincide with the vehicle centre, and different observers can track different surfaces of the same car. Extended-object tracking explicitly accounts for the relationship between object motion, physical extent, and multiple detections [10].

With several observers, the association problem extends from detections to tracks. Although a local track label identifies an observation history within one source, track-to-track association must determine whether histories from different sources describe the same physical object. Establishing that correspondence and fusing the associated state estimates remain distinct estimation problems [11].

C. Traffic Conflict Measures

Traffic-conflict measures quantify proximity to a collision under specified motion assumptions. Time to collision (TTC) is the time until the first predicted contact if those assumptions continue to hold [12]. For two vehicles following a straight path at constant speeds, with follower speed $v_f > v_l$ and leader speed v_l, TTC and the following time gap are

$$\mathrm{TTC} = \frac{d}{v_f - v_l}, \qquad \tau_{\mathrm{gap}} = \frac{d}{v_f}, \tag{3}$$

where $d > 0$ is the bumper-to-bumper clearance. The latter is a net time gap; gross time headway instead uses front-to-front spacing. These measures describe different properties: two vehicles travelling at the same speed can have a short following gap while their constant-speed TTC is infinite. Close following and an imminent collision are therefore distinct conditions [13].

For cut-ins and crossing traffic, the distance-to-closing-speed ratio alone does not establish a collision course. Predictions must account for both vehicles' paths and physical footprints. Measures based on braking further consider the deceleration needed to avoid predicted contact after an assumed response delay. Their interpretation depends on the motion model, response assumptions, and available deceleration [14].

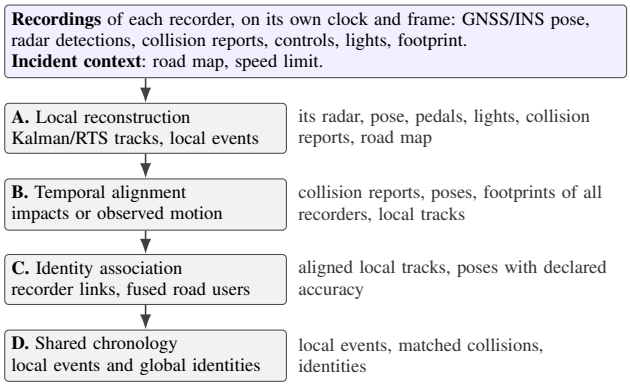


Fig. 2. Reconstruction stages and inputs. Local events are extracted independently before temporal and identity integration.

They characterize aspects of conflict urgency without establishing its cause.

The interpretation of response timing likewise depends on context. Naturalistic studies show that braking onset varies with the evolving kinematic urgency of an encounter, rather than following a universal delay after a single hazard onset [15]. An interval from a conflict observable in sensor data to brake application therefore describes an observed braking delay; it does not establish when the driver perceived the hazard or whether the driver was distracted.

III. Methodology

The method combines independent onboard recordings to reconstruct a collision retrospectively. Each recording vehicle, or recorder, contributes a GNSS/INS pose stream with declared accuracy, control and indicator logs, radar detections, and collision reports, all on its own clock. Radar supplies range, azimuth, elevation, and radial velocity without identity, while an impact report supplies contact time and impulse magnitude without naming the partner. Each recorder also knows its footprint and sensor mounts; the investigation supplies the road map and speed limit. Radar observations from the recorders provide the evidence for reconstructing road users without accessible recordings. The four stages of Fig. 2 read only the inputs shown there.

A. Local Reconstruction

Local processing retains each recorder's clock and uses a frame anchored at its initial pose. Within this frame, sensor mounting geometry and recorded ego-motion transform radar positions and compensate Doppler, including the radar mount's lever arm. Tracks follow the constant-velocity model of (1) and are updated with the median position of associated returns and their Doppler measurements. Before a detection can update a track, it must pass both spatial and radial-velocity gates. The position innovation uses (2), with covariance

$$\begin{aligned} \mathbf{S} &= \mathbf{P}_{pp} + 2\mathbf{B} + \mathbf{R}(\phi)\,\mathrm{diag}(\sigma_\rho^2, \rho^2\sigma_\phi^2)\mathbf{R}(\phi)^\mathsf{T}, \\ \mathbf{B} &= \mathbf{R}(\hat{\theta})\,\mathrm{diag}(L^2/12, W^2/12)\mathbf{R}(\hat{\theta})^\mathsf{T}, \end{aligned} \tag{4}$$

where $\mathbf{P}_{pp}$ is the predicted position covariance and $\mathbf{R}(\alpha)$ is a planar rotation. The radar term depends on range ρ, bearing ϕ, and their measurement standard deviations, while $\mathbf{B}$ models a uniform point over a nominal $L \times W$ vehicle at heading $\hat{\theta}$, with isotropic spread when heading is unavailable. The factor two accounts for the return and tracked point lying at different positions on that body. Consequently, the gate widens laterally with range, while Doppler constrains only line-of-sight velocity.

When several tracks pass both gates, their spatial and Doppler likelihoods decide the assignment. A detection without a clear likelihood margin updates no track and starts none, avoiding a forced association. New tracks require sustained support and are suppressed within a confirmed body's region. Track continuation ends when prediction uncertainty prevents separation of vehicle-width bodies, subject to a gap cap. To accommodate abrupt manoeuvres and impacts, the affected tracks' motion uncertainty is increased. RTS smoothing [8] is then applied within segments separated by gaps or detected impacts, preserving velocity discontinuities.

B. Temporal Alignment

Temporal alignment brings these independent local reconstructions onto a common clock. Collisions provide the main anchors because a contact between two recorders appears in both reports, with consecutive sensor callbacks grouped into local contact episodes. Reports from different recorders are matched on impulse magnitude: the two bodies receive equal and opposite impulses, so the magnitudes must agree within their measurement uncertainty. Impulse agreement alone, however, cannot distinguish unrelated contacts of similar strength. A candidate match must therefore also satisfy geometric consistency: placed by their own pose streams, the two footprints must touch within the declared positioning uncertainty. Where available, pre-contact radar motion must not contradict the candidate partner's recorded speed, and every candidate must agree with the clock offsets already established. If these checks still permit more than one pairing, a match is accepted only when every feasible one-to-one assignment with the greatest number of matches contains it. Each accepted match constrains relative clock offsets, while chains of matched collisions extend the alignment to recorders that never collided with each other. The strongest matched collision defines the global time origin, giving $t_\mathrm{g} = t + o_R$ for recorder R, where o_R is its estimated constant offset. The constant-offset model assumes recorder clocks disciplined by GNSS time, so that drift within one incident can be neglected.

For a recorder without a shared collision, radar observations of recording vehicles may still provide alignment. Candidate offsets are scored by comparing the vehicle surfaces seen by radar with the footprint and trajectory of the corresponding recorder, within the measurement uncertainty. The best fit is accepted only when it is consistent and distinguishable from competing offset intervals; otherwise the recorder remains unaligned. When no recorder shares a collision at all, one recorder's clock is taken as the reference.

C. Identity Association

With the recorders on a common clock, their tracks can be compared against the known trajectories of the recorders

themselves. These are the first association candidates; each radar track keeps its local identifier until it is associated with a road user. A track observed by recorder X is associated with recorder Y when its reconstructed position and line-of-sight velocity agree with Y's recorded trajectory and vehicle footprint. This comparison accounts for radar noise, the declared positioning accuracy of both recorders, and the displacement that a heading error causes at range. Linking requires agreement that lasts long enough and is precise enough to distinguish nearby vehicles, with a clear likelihood margin when two matches compete. Tracks approaching a matched collision provide further association evidence. When positioning uncertainty is declared, observations of recorder footprints refine relative translation offsets before association and fusion are repeated; unconstrained components retain their positioning priors.

Road users without a recorder have no trajectory of their own against which observations can be checked, so reconstruction instead associates the tracks of different observers with each other. Direct comparison of their tracked points would be misleading because each observer sees a different face of the body. The method therefore tests whether one vehicle body explains both sets of radar returns. For an estimated heading, returns are assigned to faces visible from their radar origins; constraints normal to and along each face then enter a weighted least-squares fit of the centre, length and width. Measurement and pose uncertainty determine the constraint weights, while nominal dimensions provide size priors. Face assignment and fitting are performed iteratively. The χ^2 terms below are the resulting weighted residual sums, including the dimension priors. At each shared observation, the compatibility cost is

$$D_{\mathrm{body}} = \max\bigl(0, \chi^2_{\mathrm{joint}} - \chi^2_a - \chi^2_b\bigr)\,, \tag{5}$$

where the terms are the normalized fitting costs for one joint body and two separate bodies. This cost is tested against a χ^2_2 gate, together with velocity compatibility; after either observer's impact, geometry alone is used. Agreement must persist, and a link is refused when a rival remains plausible or a *cannot-link* constraint would be violated: tracks that one observer simultaneously distinguishes as separate bodies cannot be fused.

Across short gaps, track continuation is evaluated against rival tracks and a new-road-user hypothesis. Retrospective resolution requires a later group supported by sustained cross-observer evidence: its first measured position and velocity are propagated backward with the motion model and compared with earlier fragments' last measured states, including process and relative-pose uncertainty. A candidate must pass a χ^2_4 state-compatibility gate and reach model posterior support of at least 0.95 against both competing successors and competing predecessors, with a new-road-user alternative in each comparison. Merging also requires that the cannot-link constraints hold and that the earlier fragment ends without a confusable neighbour. Successful resolution updates identity attribution while preserving the original observations and event times, and records when the resolving evidence became available.

Tracks not linked to a recorder form anonymous global road users $U_1, U_2, \ldots$. Their positions come from the body fitted to the observed surfaces, and the velocity estimates of different observers are combined by covariance intersection [16], minimizing the trace of the fused covariance. This handles unknown cross-correlations and also provides the position fallback when no joint body fit is available. Fused trajectories are smoothed without crossing gaps or impacts.

D. Semantic Events and Shared Chronology

As part of local reconstruction, semantic events are extracted independently from each recorder's tracks, motion, controls, lights, and road context, ready for integration on the recovered clock in the final stage. Pedal and light logs provide braking and indicator events directly, whereas interactions are judged against the map-matched path of the recorder, following lane curvature and its recorded branch at junctions. An `EGO_PATH_ENTRY` marks the instant at which the estimated body of another road user enters this corridor. Since occupancy is computed from the reconstructed footprint with its position and heading uncertainty, a transition needs persistent evidence beyond that uncertainty rather than a single observation. Each entry is then attributed to the target's lateral motion, the recorder's own manoeuvre, or both, and this attribution separates a cut-in from a lane change of the recorder: a `CUT_IN` needs a sustained lateral approach from another lane and is confirmed only by a path entry to which the target's motion contributes.

Alongside path entries and cut-ins, which describe where other road users are, a `CRITICAL_TTC` event marks when such a situation becomes critical through either of two triggering conditions. The first is a predicted overlap between the recorder's safety envelope and another vehicle's estimated body that cannot be avoided with the assumed deceleration after a response delay [17]. Avoidance is evaluated by braking the recorder, or the target when recorder braking cannot remove the overlap. Predictions account for road curvature and estimated target braking. The second is an unsafe forward gap below a speed-dependent following-distance reference derived from the time-gap table of UN Regulation No. 157 [18]. This condition applies only to a leader in the recorder's path or a confirmed cut-in, never to traffic in adjacent lanes, and unlike a predicted overlap, a short following gap does not by itself give a time-to-collision [13]. Both conditions require sufficiently mature tracks with bounded position and velocity uncertainty. The experiments use a 1.0 s response delay, 6.0 m/s^2 available deceleration and a 6.0 s prediction horizon.

Fused trajectories describe the estimated physical motion, while the shared chronology integrates events already extracted independently by each recorder. Local semantic events, matched collisions, and associated identities form a graph on the global clock; each node keeps its recorder, local event, local timestamp, and provenance. A matched pair of collision reports becomes one shared node with both participants, while an unmatched report stays a node of its own recorder. Brief recontacts of the same pair are grouped with the preceding collision when the two footprints barely separated in between.

TABLE I
ADDED ERROR PROFILES. POSE TARGETS USE THE INDICATED PERCENTILES. RADAR PARAMETERS ARE GAUSSIAN STANDARD DEVIATIONS, PER-RETURN DROP PROBABILITY p, AND MEAN FALSE RETURNS PER SWEEP PER SENSOR λ; $\sigma_\angle$ APPLIES TO BOTH AZIMUTH AND ELEVATION.

Pose target	**Position [m]**		**Heading [°]**	**Velocity [m/s]**	
G1 (P95)	0.040		0.140	0.016	
G2 (P99.5)	0.500		0.600	0.200	
Radar	σ_ρ [m]	$\sigma_\angle$ [°]	σ_v [m/s]	p	λ
Nominal	0.25	0.5	0.10	0.20	3
Severe	0.50	2.0	0.20	0.40	3
Extreme	0.50	2.0	0.20	0.40	30

Temporal edges then order events with different reconstructed timestamps, leaving events that share a timestamp unordered.

IV. EVALUATION

We evaluate the reconstruction of physical identities and semantic events from unperturbed recordings before examining its sensitivity to controlled positioning and radar errors. We use the same recordings and reconstruction settings in both parts.

A. Experimental Setup

The dataset contains 19 scripted scenarios and 22 runs acquired in CARLA 0.9.15 at 20 Hz [19], [20]. The scenarios span three maps and cover rear-end, cut-in, junction and chain collisions, occlusions, dense traffic and motorway driving. Across the recordings, 50 recording and 45 non-recording vehicles are present, with all 26 evaluated collisions occurring between recording vehicles. Each recorder supplies GNSS/INS poses, controls and indicators, impact reports without partner identities, and three radar streams: a front radar with a 150° field of view and two side radars with 140°, each with 90 m range and 600 rays per sweep, and no rear coverage. G0 uses exact poses and native radar returns without added corruption, although field of view, occlusion and finite ray sampling still limit observations. By leaving impact-report streams unchanged, the campaign tests collision matching under positioning and radar errors rather than crash-sensor failures.

The position and heading perturbations in Table I combine constant bias, a Gauss–Markov component and white noise, while velocity errors are white. G1 follows measured RTK-INS accuracy [21], and G2 uses stronger targets informed by field trials and receiver specifications [22], [23]. The radar profiles combine Gaussian measurement errors, independent return losses and Poisson clutter, with values drawn from automotive measurements, published models and engineering stress choices [24]–[26]. False-return locations are uniform over range and field of view, unlike structured real clutter [27]; the extreme clutter mean serves as a model stress value, not an estimated deployment frequency.

The experimental campaign comprises 1816 jobs covering G0, positioning-only errors, radar errors and combined conditions. All-recorder radar profiles affect every radar on every recording vehicle throughout the log, while asymmetric jobs enumerate selected recorders or subsets. Corresponding jobs use identical per-recorder perturbations, but the pooled scenario weights differ: single-recorder rows contain one job per recorder and all-recorder rows contain one per run. Stochastic conditions use three seeds, or five for extreme radar. Reconstruction results are averaged over seeds, and selected paired changes include 95% Student's t intervals over those seeds to quantify perturbation variability on the designed cases.

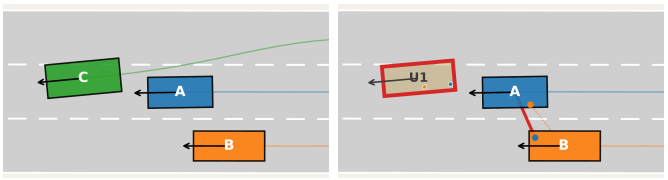
Ground truth enters only at scoring, where physical association is evaluated through correct, wrong and missed recorder links, entity purity, false merges and fragmentation, independently of arbitrary identifier names. Measured track samples are assigned to the nearest true vehicle box within 1.5 m. Entity purity then measures the share carrying the most frequent assignment, while an unmatched mode defines a phantom. Position errors concern non-recording entities at states where the contributing tracks agree with the same true vehicle, with impure states, false entities and recorder duplicates counted separately.

To assess whether semantic conversion preserves the manoeuvres and conflict timing needed to interpret the recorded sequence, local scoring covers newly emitted path-entry, left/right cut-in and `CRITICAL_TTC` onsets. Predictions and oracle events are restricted to before each recorder's first contact *before* one-to-one matching by type, physical subject and time within 2 s; witnesses without contact contribute their full recording. Duplicate onsets count as false positives, and precision and radar-observable recall pool these counts across jobs. Since physical subjects are assigned from truth correspondence, local event scores complement, rather than replace, global association measures. The last-second diagnostic captures unsupported context near impact by counting jobs with a nonduplicate unmatched conflict interval overlapping $[t_c - 1, t_c)$ before at least one recorder's first contact.

B. Reconstruction Across the Scenarios

Under G0, all 26 collision participant sets are correct and all 45 observable non-recording vehicles are represented, with median position error 0.24 m (P95: 0.82 m) and median speed error 0.04 m/s. The physical association results show correct recorder links in 49 cases, none wrong and one missed. No false merge occurs, and measured-sample purity is 1.0 for 64 of 66 anonymous entities, with twenty duplicate entities and one recorder duplicate counted separately. Together, these measures assess consistency of physical identity independently of agreement between local and global label names.

To assess the geometric reconstruction under G0, we compare the fitted bodies with nominal boxes inferred independently from each source track. The comparison pools 4376 source-consistent entity instants across all 45 observable non-recording vehicles in the ten eligible runs. At each instant, the reference is the median centre error over available source boxes, and instants without a reference are excluded from both sides. On this matched population, the combined body-reconstruction and fusion step reduces median error from 0.484 to 0.240 m and P95 from 1.334 to 0.806 m. The



(a) Ground truth (b) Reconstruction

Fig. 3. S12, one second before contact. C neither collides nor supplies a recording. Its radar observations from A and B form U1; red overlays show A's reconstructed conflicts with U1 and B.

median and P95 decrease in all ten runs, while individual errors decrease at 81.4% of paired instants.

Local conflict precision is 0.978 and observable recall 0.899 (Table II), with a median matched onset error of 0.05 s. The sequence recovered in S02 instantiates the running example: B enters A's path just outside the following-distance reference, after which the gap becomes unsafe and B's braking makes the predicted encounter braking-critical. A's recorded brake command follows the first reconstructed critical onset by 2.60 s, while cut-in, path-entry and critical onsets agree with the oracle within 0.10 s. The chronology thus recovers the gap left by B and the braking sequence that the final contact alone does not preserve.

Fig. 3 shows how the reconstruction includes a vehicle absent from the impact reports. In S12, C cuts into A's lane before A swerves into B's lane and the two recorders collide. A's local evidence already contains C's cut-in; association brings three tracks from A and B into one road user, U1, with purity 1.0. By placing U1's cut-in and path entry before A's critical interaction with B, the shared chronology preserves both observers' evidence of the preceding interaction without requiring a recording from C.

C. Robustness to Sensor Degradation

Positioning errors under G1 and G2 preserve correct identification of every collision and representation of every observable non-recording vehicle, with neither wrong recorder links nor false merges. The paired changes in road-user P95 are also small: G1 adds 0.031 m (95% CI: $[-0.027, 0.089]$ m), and G2 adds 0.050 m ($[-0.075, 0.175]$ m). Nominal radar likewise preserves collision identification, with position P95 below 1 m. Position errors, false entities and association errors increase under severe and extreme radar, although the all-recorder tests retain 98.7% and 95.4% correct collision participant sets, respectively. Across the entire campaign, all 2344 true contact instances are detected and 2322 have correct participant sets; 22 extra single-recorder impact nodes are also produced. Reporting these outcomes separately keeps collision identification distinct from radar identities and semantic correctness.

In the asymmetric rows, severe corruption of one recorder retains 99.5% correct collision sets, while extreme corruption reduces local event precision to 0.509. Adding G2 to all-recorder severe radar changes position P95 little but reduces local precision from 0.846 to 0.826. The sustained all-recorder profiles, which leave no unaffected radar observer, characterize stress sensitivity rather than the frequency of simultaneous failures.

Two supplementary tests perturb the clocks and the impact reports, using three seeds per run under G0 and G2 with severe radar. Clock offsets drawn uniformly within ± 50 ms per recorder are recovered within 0.1 ms for successfully aligned recorders; one of 150 recorders loses radar-based alignment under G2 with severe radar. Reconstructed event times nevertheless shift by up to 0.3 s under G0 and 0.75 s under G2 with severe radar, reversing 0.25% and 0.40% of the evaluated event-pair orderings, respectively. EDR-inspired impact reports are perturbed with ΔV noise having 5% or 10% standard deviation per impact, compared with the $\pm 10\%$ measurement accuracy specified in US regulations [28], together with 5% mass error per recorder and up to ± 10 ms of timing error. Collision participant sets remain correct in all 264 impact-perturbation jobs.

D. Reliability of the Event Chronology

For the surrounding chronology, positioning-only and nominal-radar conditions retain local conflict precision above 0.95 and observable recall near 0.90, accompanied by a median matched onset error of 0.05 s and median matched interval overlap of about 0.97. With one severely degraded recorder, precision is 0.911 and 8.7% of jobs contain an unmatched nonduplicate interval in the last second before a recorder's first contact; with all recorders severely degraded, these values become 0.846 and 15.2%.

The distinction between collision identification and contextual evidence becomes more pronounced under extreme radar. With every recorder affected, precision falls to 0.313 while observable recall remains 0.834, and 793 of 907 false predictions concern phantom subjects. Median matched onset error remains 0.10 s, but this timing accuracy applies to matched events and does not validate the unmatched ones. Last-second unmatched intervals occur in 86.4% of jobs, and even one extremely degraded recorder gives 56.8%. The chronology therefore preserves many genuine conflicts while adding substantial unsupported context under dense clutter. Its claims remain auditable through recorder provenance, original local events and retrospective identity evidence, with their reliability assessed through the separate association and event-quality measurements.

V. Related Work

Reconstructing a collision from vehicle recordings connects accident investigation with cooperative perception and multi-sensor estimation. We consider how these approaches recover motion and relate vehicle identities and recordings in time to support forensic interpretation.

A. Accident Reconstruction from Recorded Evidence

Dynamics models provide a basis for reconstructing vehicle motion and contacts in simulation tools such as PC-Crash [31], with EDR measurements offering independent checks on the resulting estimates [32]. Recorded perception and exchanged

TABLE II

SELECTED CONDITIONS. RECONSTRUCTION COLUMNS ARE SEED MEANS; LOCAL EVENT SCORES ARE POOLED COUNTS. COLL.: FRACTION OF CORRECTLY IDENTIFIED CONTACTS; PHANT., GHOSTS, W. LINKS AND F. MERGES: COUNTS PER JOB, WITH GHOSTS BEING RECORDERS DUPLICATED AS ANONYMOUS ENTITIES; P95: SOURCE-CONSISTENT ROAD-USER POSITION ERROR. PREC./REC.: PHASE-RESTRICTED ONSET PRECISION AND OBSERVABLE RECALL, INCLUDING DUPLICATE ONSETS AS FALSE POSITIVES. LAST 1 S: PERCENTAGE OF JOBS WITH AN UNMATCHED NONDUPLICATE INTERVAL OVERLAPPING THE SECOND BEFORE A RECORDER'S FIRST CONTACT.

		Reconstruction						Local conflict events		
Condition	**Jobs**	**Coll.**	**Phant.**	**Ghosts**	**W. links**	**F. merges**	**P95 [m]**	**Prec.**	**Rec.**	**Last 1 s**
G0	22	1.000	0.0	0.05	0.00	0.00	0.82	0.978	0.899	4.5%
G1, RTK	66	1.000	0.0	0.05	0.00	0.00	0.85	0.982	0.902	4.5%
G2, degraded	66	1.000	0.0	0.02	0.00	0.00	0.87	0.986	0.906	3.0%
Radar nominal	66	1.000	0.6	0.06	0.00	0.00	0.97	0.954	0.902	7.6%
Radar severe	66	0.987	68.7	0.50	0.03	0.68	1.33	0.846	0.902	15.2%
Radar extreme	110	0.954	102.4	1.79	0.12	0.91	1.56	0.313	0.834	86.4%
One recorder, severe	150	0.995	30.5	0.26	0.01	0.03	1.08	0.911	0.892	8.7%
One recorder, extreme	250	0.981	45.6	0.84	0.05	0.06	1.19	0.509	0.862	56.8%
G2 + radar severe	66	1.000	69.6	0.52	0.03	0.73	1.33	0.826	0.879	25.8%

TABLE III

CLOSEST FORENSIC RECONSTRUCTION SYSTEMS. ENTRIES SUMMARIZE THE REPORTED PROCEDURES AND DEMONSTRATIONS.

Work	**Recorded evidence**	**Temporal alignment**	**Reconstruction and interpretation**
Beck et al. [3]	Camera, LiDAR, radar; vehicle logs	Synchronized inputs; recovery outside scope	Image/depth tracking of a selected target
Matsumura et al. [4]	EDR + dashcam from one or both vehicles	Chronological EDR/video integration	Trajectories of known crash participants
Zhang et al. [29]	Driving logs + dashcam/map	Vehicle-log timestamps	Motion from logged relative distances
Yuan et al. [30]	EDR + GPS + roadside video	External time reference and correction	Kalman filtering; vehicle-state and scene-parameter fusion
Ours	Raw radar + vehicle logs	Clock recovery from shared contacts or observed motion	Cross-observer association; Kalman filtering and RTS smoothing; semantic events with recorder provenance

vehicle states provide complementary routes to motion reconstruction. DeepCrashTest uses trajectories extracted from monocular dashcam videos to create virtual crash tests [33], while Young et al. reconstruct host and neighbouring trajectories from an EDR storing states received through cooperative collision-warning messages [34].

B. Cooperative Perception and Tracking

Combining communicated object tracks in high-level cooperative perception [35] requires correspondences between vehicles' object lists [36] as well as spatial and temporal alignment [37]. Differences in observation time are addressed through asynchronous automotive track fusion by Aeberhard et al. [38], time-compensated late fusion in DAIR-V2X [39], and cooperative detection under asynchronous sharing and pose errors in V2X-ViT [40].

Uncertainty is addressed by Chiu et al., who learn detection uncertainty within a multisensor Kalman framework for cooperative tracking [41], while Castelino et al. compare track-fusion strategies using Collective Perception Messages in asynchronous scenarios, assuming reliable source timestamps [42]. Together, these works establish asynchronous fusion and cross-vehicle association as central tasks in cooperative perception.

C. Association, Fusion, and Temporal Registration

Correlated estimation errors affect both track association and fusion [43], [44]; covariance intersection provides an established way to combine estimates when their cross-correlations are unknown [16]. For automotive radar, extended-object tracking also models the multiple reflection centres on one vehicle [9], while consistent cooperative localization addresses uncertainty in relative pose and correlation [45]. These principles inform our association tests, which ask whether one rigid body can explain the surfaces seen by different observers while accounting for uncertainty and competing identities. Later evidence can then establish earlier identities while preserving the original local observations and event times.

Common scene dynamics [46] and passive sensor timing estimation [47] likewise provide precedents for temporal registration. In the present method, shared collision reports constrain constant clock offsets after checks against recorded geometry and available radar motion. When this evidence cannot align a recorder, its offset is estimated by comparing radar observations of a recording vehicle with that vehicle's own logged motion and footprint. Temporal registration is thus coupled to the physical evidence used for vehicle association.

D. Forensic Timelines and Interpretation

The NTSB's Tempe investigation demonstrates the forensic value of relating an automated vehicle's perception history to its subsequent response [48]. Response models relate action timing to the evolving situation and driver expectations [49], while counterfactual studies such as Scanlon et al.'s reconstructed fatal crashes assess alternative outcomes through

additional simulation [50]. These applications motivate preserving the connection between reconstructed interactions and recorded responses in a forensic chronology.

E. Positioning of This Work

Table III compares how the closest forensic systems turn recorded evidence into aligned motion and reconstruction outputs. Image and depth tracking of a selected target allow Beck et al. [3] to reconstruct uninstrumented surrounding vehicles, with clock synchronization outside their scope. Other systems integrate vehicle records with scene information: Matsumura et al. [4] combine EDR trajectories and dashcam frames from one or both crash participants in a chronological road map, while Zhang et al. [29] reproduce traffic scenes in OpenSCENARIO from timestamped driving logs and relative distances. Explicit temporal alignment is central to Yuan et al. [30], who align EDR, GPS and roadside video to an external time reference and use Kalman filtering to handle noise and timing deviations.

In our setting, both clock offsets and physical vehicle correspondences must be recovered across independent radar recordings. Shared contacts and observed motion constrain relative time, and body geometry connects the different surfaces seen by different observers. Motion is estimated through Kalman filtering and RTS smoothing, while sustained cross-observer evidence can also resolve earlier identity fragments. Together, these steps support reconstruction of vehicles with and without their own logs, including road users absent from the impact reports, without requiring camera-based correspondence or communication during the incident.

The semantic chronology makes this reconstruction inspectable across recorders by relating manoeuvres and conflicts to recorded responses and contacts, while retaining local observations, event times and the provenance of later identity decisions. The contribution therefore lies in integrating temporal and physical correspondence recovery with retrospective motion estimation and a traceable semantic account. The evaluation examines these aspects separately through collision identification, physical association, geometry and local event fidelity under controlled sensor degradation.

VI. Threats to Validity

Validity depends on correct temporal and identity correspondences and on interpreting the chronology within the reconstructed evidence. We address these risks through controls in the reconstruction and separate evaluation of its outputs.

a) Internal validity: Because an incorrect impact pairing could propagate clock and identity errors through the chronology, matching checks impulse compatibility, footprint contact, available radar motion and consistency with established offsets. Reconstruction uses recorder data and supplied road context, with ground truth entering only at scoring. To reduce confounding from configuration changes or different noise realizations, the campaign fixes recordings, code and settings and reuses identical per-recorder perturbations in corresponding conditions.

b) External validity: Transfer to other sensing conditions depends on the structure of the errors as well as their magnitude. The severe and extreme all-recorder profiles apply sustained corruption to every radar throughout every recording, while asymmetric runs assess partial degradation. Both quantify sensitivity to the specified conditions without estimating their deployment frequency. The error structure matters for persistent radar ghosts, which could accumulate support that independently redrawn clutter does not. Confirmation, Doppler and cross-observer body checks constrain unsupported hypotheses, although resilience to coherent ghosts satisfying these checks remains unmeasured. The collision results are obtained with impact reports available, while positioning and radar evidence are perturbed.

c) Construct validity: The oracle tests recovery of the defined manoeuvre and conflict onsets by applying the same event rules to true trajectories. Criticality still depends on the stated braking, response-delay and following-distance assumptions. Since smoothing and manoeuvre confirmation use later observations, semantic onsets are retrospective estimates of those conditions. Relating them to recorded commands while preserving source observations and identity provenance supports an auditable interpretation, without equating a reconstructed conflict with driver awareness or responsibility.

d) Conclusion validity: Because the 1816 jobs repeat 22 recorded runs, seed-based intervals describe perturbation variability within that set. Pooled differences between single-recorder and all-recorder rows are interpreted descriptively because their scenario weights differ. Local event scoring uses ground-truth subject correspondence, whereas recorder links, purity and merges separately assess physical identity. Likewise, position errors are conditional on source-consistent states, with separate false-entity and association counts exposing excluded errors. The G0 geometric comparison pairs the same entity instants and measures body reconstruction and fusion together. Reporting these dimensions separately makes the reliability of the surrounding chronology visible alongside collision identification.

VII. Conclusion

We presented a method that reconstructs road collisions from independent onboard recordings by connecting evidence across recorders through clock recovery, physical vehicle association and retrospective motion estimation. This includes observations of vehicles without their own logs. Through semantic conversion, the resulting chronology relates manoeuvres and conflicts to recorded control actions and contacts while keeping source observations and the evidence behind identity decisions traceable.

In the reference condition, the method correctly identified all 26 collision participant sets and represented all 45 observable non-recording vehicles across 19 simulated scenarios. Across the 1816-job evaluation, tested positioning and nominal radar errors preserved collision identification and high local conflict precision. When every recorder underwent extreme radar corruption, 95.4% of collision participant sets remained correct while local conflict precision fell to 31.3%, revealing the limit

of semantic interpretation under dense clutter. This chronology supports investigators in examining competing accounts of a collision.

Future work will extend validation to real onboard recordings and examine how motion and identity uncertainty can inform confidence estimates for semantic events under dense radar clutter. We will also investigate large language models for abductive reasoning over kinematic and semantic traces, generating and comparing candidate collision explanations against supporting and contradicting recorded evidence.

REFERENCES

[1] Gauthier & Maier Law Firm, P.C., "3 car accident scenarios: Who's at fault? who pays?" Jan. 2026, accessed: Oct. 28, 2026. [Online]. Available: https://www.maierattorney.com/3-car-accident-scenarios-whos-at-fault-who-pays

[2] Davis Kelin Law Firm LLC, "Handling a car accident where fault is unclear," 2024, accessed: Oct. 28, 2026. [Online]. Available: https://daviskelin.com/car-accidents/handling-a-car-accident-where-fault-is-unclear/

[3] J. Beck, R. Arvin, S. Lee, A. Khattak, and S. Chakraborty, "Automated vehicle data pipeline for accident reconstruction: New insights from LiDAR, camera, and radar data," *Accident Analysis & Prevention*, vol. 180, p. 106923, 2023.

[4] H. Matsumura, M. Sugiyama, and T. Iwata, "Study on a method for reconstructing pre-crash situations using data of an event data recorder and a dashboard camera," SAE International, SAE Technical Paper 2024-01-2891, 2024.

[5] L. Bacchiani, A. Bedei, M. Bravetti, R. Girau, and G. Pau, "CausalDrive Forensics: Code, recordings, and results," Software repository, 2026, reconstruction frozen at tag campaign-v3. [Online]. Available: https://github.com/AndreaBedei1/causaldrive-forensics

[6] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation: Theory, Algorithms and Software*. New York, NY, USA: Wiley, 2001.

[7] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[8] H. E. Rauch, F. Tung, and C. T. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, vol. 3, no. 8, pp. 1445–1450, 1965.

[9] L. Hammarstrand, F. Sandblom, L. Svensson, and J. Sörstedt, "Extended object tracking using a radar resolution model," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 48, no. 3, pp. 2371–2386, 2012.

[10] K. Granström, M. Baum, and S. Reuter, "Extended object tracking: Introduction, overview and applications," *Journal of Advances in Information Fusion*, vol. 12, no. 2, pp. 139–174, 2017.

[11] C.-Y. Chong, S. Mori, W. H. Barker, and K.-C. Chang, "Architectures and algorithms for track association and fusion," *IEEE Aerospace and Electronic Systems Magazine*, vol. 15, no. 1, pp. 5–13, 2000.

[12] J. C. Hayward, "Near-miss determination through use of a scale of danger," *Highway Research Record*, no. 384, pp. 24–34, 1972.

[13] K. Vogel, "A comparison of headway and time to collision as safety indicators," *Accident Analysis & Prevention*, vol. 35, no. 3, pp. 427–433, 2003.

[14] L. Westhofen, C. Neurohr, T. Koopmann, M. Butz, B. Schütt, F. Utesch, B. Neurohr, C. Gutenkunst, and E. Böde, "Criticality metrics for automated driving: A review and suitability analysis of the state of the art," *Archives of Computational Methods in Engineering*, vol. 30, no. 1, pp. 1–35, 2023.

[15] G. Markkula, J. Engström, J. Lodin, J. Bärgman, and T. Victor, "A farewell to brake reaction times? kinematics-dependent brake response in naturalistic rear-end emergencies," *Accident Analysis & Prevention*, vol. 95, pp. 209–226, 2016.

[16] S. J. Julier and J. K. Uhlmann, "A non-divergent estimation algorithm in the presence of unknown correlations," in *Proc. American Control Conference (ACC)*, vol. 4, 1997, pp. 2369–2373.

[17] M. Green, ""how long does it take to stop?" methodological analysis of driver perception-brake times," *Transportation Human Factors*, vol. 2, no. 3, pp. 195–216, 2000.

[18] United Nations Economic Commission for Europe, "UN regulation no. 157: Uniform provisions concerning the approval of vehicles with regard to automated lane keeping systems," Official Journal of the European Union, L 82, p. 75, 2021, 01 series of amendments: ECE/TRANS/WP.29/2022/59/Rev.1.

[19] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proc. 1st Annual Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 78, 2017, pp. 1–16.

[20] CARLA Team, "CARLA 0.9.15 documentation: Sensors reference," https://carla.readthedocs.io/en/0.9.15/, accessed: Oct. 2026.

[21] M. Volckaert, D. Schellekens, K. Smolders, A. Simsky, and B. Bougard, "Integrity of an RTK-INS positioning system using SSR corrections for safety-critical automotive applications," Septentrio technical presentation, https://gnss-imu.com/down/upload/20240402/1712021876.pdf, Sep. 2019, accessed: Oct. 2026.

[22] R. Lesjak *et al.*, "Centimeter-level GNSS positioning using C-ITS for correction data delivery: An experimental study," in *Lecture Notes in Mobility*. Cham, Switzerland: Springer, 2025, pp. 565–571.

[23] u-blox, "ZED-F9R-02B data sheet," Document UBX-21017486, rev. R06, May 2024.

[24] O. Schumann *et al.*, "RadarScenes: A real-world radar point cloud data set for automotive applications," in *Proc. IEEE International Conference on Information Fusion (FUSION)*, 2021, pp. 1–8.

[25] A. Scheel and K. Dietmayer, "Tracking multiple vehicles using a variational radar model," *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 10, pp. 3721–3736, 2019.

[26] F. Kraus, N. Scheiner, W. Ritter, and K. Dietmayer, "The radar ghost dataset: An evaluation of ghost objects in automotive radar data," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 8570–8577.

[27] J. Kopp, D. Kellner, A. Piroli, and K. Dietmayer, "Fast rule-based clutter detection in automotive radar data," in *Proc. IEEE International Intelligent Transportation Systems Conference (ITSC)*, 2021, pp. 3010–3017.

[28] National Highway Traffic Safety Administration, "49 CFR part 563: Event data recorders," https://www.ecfr.gov/current/title-49/subtitle-B/chapter-V/part-563, accessed: Oct. 2026.

[29] S. Zhang, X. Li, Z. Zha, H. Xiong, C. Wu, P. Li, and Z. Kang, "A study on in-depth accident investigation of intelligent connected vehicles based on reverse scenario construction," *Scientific Reports*, vol. 15, p. 44969, 2025.

[30] Q. Yuan, W. Ji, R. Cheng, S. Ye, T. Wang, and T. Cui, "Multisource data alignment and fusion for traffic accident reconstruction: Integrating EDR, GPS, and video recordings," *Digital Transportation and Safety*, vol. 5, no. 3, pp. 229–236, 2026.

[31] H. Steffan and A. Moser, "The collision and trajectory models of PC-CRASH," SAE International, SAE Technical Paper 960886, 1996.

[32] H. C. Gabler, C. E. Hampton, and J. Hinch, "Crash severity: A comparison of event data recorder measurements with accident reconstruction estimates," SAE International, SAE Technical Paper 2004-01-1194, 2004.

[33] S. K. Bashetty, H. Ben Amor, and G. Fainekos, "DeepCrashTest: Turning dashcam videos into virtual crash tests for automated driving systems," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 11 353–11 360.

[34] C.-P. Young, B.-R. Chang, and T.-Y. Wei, "Highway vehicle accident reconstruction using cooperative collision warning based motor vehicle event data recorder," in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2009, pp. 1131–1136.

[35] A. Rauch, F. Klanner, R. Rasshofer, and K. Dietmayer, "Car2X-based perception in a high-level fusion architecture for cooperative perception systems," in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2012, pp. 270–275.

[36] A. Rauch, S. Maier, F. Klanner, and K. Dietmayer, "Inter-vehicle object association for cooperative perception systems," in *Proc. IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2013, pp. 893–898.

[37] C. Allig and G. Wanielik, "Alignment of perception information for cooperative perception," in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2019, pp. 1849–1854.

[38] M. Aeberhard, S. Schlichtharle, N. Kaempchen, and T. Bertram, "Track-to-track fusion with asynchronous sensors using information matrix fusion for surround environment perception," *IEEE Transactions on Intelligent Transportation Systems*, vol. 13, no. 4, pp. 1717–1726, 2012.

[39] H. Yu, Y. Luo, M. Shu, Y. Huo, Z. Yang, Y. Shi, Z. Guo, H. Li, X. Hu, J. Yuan, and Z. Nie, "DAIR-V2X: A large-scale dataset for vehicle-infrastructure cooperative 3D object detection," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 21 329–21 338.

[40] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, "V2X-ViT: Vehicle-to-everything cooperative perception with vision transformer," in *Proc. European Conference on Computer Vision (ECCV)*, 2022, pp. 107–124.

[41] H.-K. Chiu, C.-Y. Wang, M.-H. Chen, and S. F. Smith, "Probabilistic 3D multi-object cooperative tracking for autonomous driving via differentiable multi-sensor kalman filter," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 18 458–18 464.

[42] R. M. Castelino, S. Pradhan, and A. Hahn, "Track-to-track fusion for cooperative perception using collective perception messages," *Sensors*, vol. 26, no. 6, p. 2003, 2026.

[43] Y. Bar-Shalom, "On the track-to-track correlation problem," *IEEE Transactions on Automatic Control*, vol. 26, no. 2, pp. 571–572, 1981.

[44] Y. Bar-Shalom and L. Campo, "The effect of the common process noise on the two-sensor fused-track covariance," *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-22, no. 6, pp. 803–805, 1986.

[45] E. Héry, P. Xu, and P. Bonnifait, "Consistent decentralized cooperative localization for autonomous vehicles using LiDAR, GNSS, and HD maps," *Journal of Field Robotics*, vol. 38, no. 4, pp. 552–571, 2021.

[46] Y. Caspi and M. Irani, "Spatio-temporal alignment of sequences," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 11, pp. 1409–1424, 2002.

[47] E. Olson, "A passive solution to the sensor synchronization problem," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2010, pp. 1059–1064.

[48] National Transportation Safety Board, "Collision between vehicle controlled by developmental automated driving system and pedestrian, tempe, arizona, march 18, 2018," National Transportation Safety Board, Tech. Rep. NTSB/HAR-19/03, 2019.

[49] J. Engström, S.-Y. Liu, A. Dinparastdjadid, and C. Simoiu, "Modeling road user response timing in naturalistic traffic conflicts: A surprise-based framework," *Accident Analysis & Prevention*, vol. 198, p. 107460, 2024.

[50] J. M. Scanlon, K. D. Kusano, T. Daniel, C. Alderson, A. Ogle, and T. Victor, "Waymo simulated driving behavior in reconstructed fatal crashes within an autonomous vehicle operating domain," *Accident Analysis & Prevention*, vol. 163, p. 106454, 2021.



Lorenzo Bacchiani received his Ph.D. in Computer Science and Engineering from the University of Bologna in 2025. He is currently a Research Fellow at the Department of Computer Science and Engineering, University of Bologna. His interests include behavioral-based methodologies for component interaction, adaptation, and deployment, as well as digital twin technologies in the automotive domain. In particular, his work explores the development and integration of vehicle, driver and road digital twins to improve system reliability and safety.



Andrea Bedei is a Ph.D. candidate in Automotive Engineering for Intelligent Mobility at the University of Bologna, Cesena Campus. He received the M.S. degree in Computer Science and Engineering from the same University in 2025. His research interests include computer vision, machine learning, simulation frameworks for autonomous vehicle safety, and the use of virtual environments for marine robotics and underwater drone experimentation.



Mario Bravetti is a Full Professor at the Department of Computer Science and Engineering of the University of Bologna. His research spans formal methods for distributed systems, concurrency and probability theory, and applications to service-oriented and cloud computing. He has contributed to numerous international research projects, served on editorial boards of international computer science journals, and authored more than 100 peer-reviewed publications. He has held key roles in major international conferences, including CONCUR, SEFM, SAC, and QEST, as well as in several international PhD schools and scientific committees.



Roberto Girau (Senior Member, IEEE) has been an associate professor at the University of Bologna, Department of Computer Science and Engineering, since 2021. He received the M.S. degree in Telecommunication Engineering and his Ph.D. degree in electronic engineering and computer science from the University of Cagliari, Italy in 2012 and in 2017, respectively. From 2012 to 2020, he worked as researcher at the Department of Electrical and Electronic Engineering of the University of Cagliari, developing an experimental platform for the social Internet of Things. His main research areas of interest are IoT and vehicular and human digital twins with particular emphasis on safety applications.



Giovanni Pau is a full professor at the University of Bologna, Department of Computer Science and Engineering (DISI). He also serves as Technical Director at the Technology Innovation Institute of Abu Dhabi and Adjunct Faculty at the University of California, Los Angeles. Previously, Dr. Pau held the ATOS/Renault smart mobility chair of excellence as a professor at Sorbonne Université, Paris, France. He received his Laurea in Computer Science and his PhD in Computer Engineering from the University of Bologna in 1998 and 2002, respectively, and completed his "Habilitation à diriger des recherches" (HDR) at Sorbonne Université in 2018. His research interests focus on vehicular robotics and IoT. His earlier work addressed network systems, particularly vehicular ad-hoc networks (VANETs), pervasive mobile sensor systems, and the application of information-centric networking (ICN) architectures to mobility and vehicular communication. His work on vehicular networks and mobile edge has been widely cited by the community and reported on in the general media, including BBC Radio, Discovery Channel, The Guardian, Wired, Corriere della Sera, and the Italian National Television. Over the years, he has published more than 100 peer-reviewed journal and conference articles.